National Coverage Determination (NCD)

Bone (Mineral) Density Studies

150.3

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Tracking Information

Publication Number
100-3

Manual Section Number
150.3

Manual Section Title
Bone (Mineral) Density Studies

Version Number
1

Effective Date of this Version
07/01/1998

Ending Effective Date of this Version
01/01/2007

Implementation Date
07/01/1998

Description Information

Benefit Category
Bone Mass Measurement

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

Bone (mineral) density studies are used to evaluate diseases of bone and/or the responses of bone diseases to treatment. The studies assess bone mass or density associated with such diseases as osteoporosis, osteomalacia, and renal osteodystrophy. Various single or combined methods of measurement may be required to: (a) diagnose bone disease, (b) monitor the course of bone changes with disease progression, or (c) monitor the course of bone changes with therapy. Bone density is usually studied by using photodensitometry, single or dual photon absorptiometry, or bone biopsy.

Indications and Limitations of Coverage

The Following Bone (Mineral) Density Studies Are Covered Under Medicare

A - Single Photon Absorptiometry

A non-invasive radiological technique that measures absorption of a monochromatic photon beam by bone material. The device is placed directly on the patient, uses a low dose of radionuclide, and measures the mass absorption efficiency of the energy used. It provides a quantitative measurement of the bone mineral of cortical and trabecular bone, and is used in assessing an individual's treatment response at appropriate intervals.

Single photon absorptiometry is covered under Medicare when used in assessing changes in bone density of patients with osteodystrophy or osteoporosis when performed on the same individual at intervals of 6 to 12 months.

B - Bone Biopsy

A physiologic test which is a surgical, invasive procedure. A small sample of bone (usually from the ilium) is removed, generally by a biopsy needle. The biopsy sample is then examined histologically, and provides a qualitative measurement of the bone mineral of trabecular bone. This procedure is used in ascertaining a differential diagnosis of bone disorders and is used primarily to differentiate osteomalacia from osteoporosis.

Bone biopsy is covered under Medicare when used for the qualitative evaluation of bone no more than four times per patient, unless there is special justification given. When used more than four times on a patient, bone biopsy leaves a defect in the pelvis and may produce some patient discomfort.

C - Photodensitometry (radiographic absorptiometry)

A noninvasive radiological procedure that attempts to assess bone mass by measuring the optical density of extremity radiographs with a photodensitometer, usually with a reference to a standard density wedge placed on the film at the time of exposure. This procedure provides a quantitative measurement of the bone mineral of bone, and is used for monitoring gross bone change.

The Following Bone (Mineral) Density Study Is Not Covered Under Medicare

D - Dual Photon Absorptiometry

A noninvasive radiological technique that measures absorption of a dichromatic beam by bone material. This procedure is not covered under Medicare because it is still considered to be in the investigational stage.

Claims Processing Instructions

TN 1416 (Medicare Claims Processing)

Transmittal Information

Revision History

07/1997 - Section 4106 of the Balanced Budget Act of 1997 (P.L. 105-217) created a bone mass measurement benefit (§1861(rr) of the Social Security Act http://www.ssa.gov/OP_Home/ssact/title18/1861.htm). This manual section was not updated to reflect the new coverage. The conforming regulations are found at 42 CFR §410.31 (http://www.gpo.gov/nara/cfr/index.html). Effective date 7/1/1998.

Additional Information

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